

TVET CENTRES OF EXCELLENCE FOR RWANDA



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Rwanda's strategic vision and implications for TVET

Rwanda's Vision 2050 emphasises that the country's ambitious economic goals can only be achieved through structural transformation. In the medium term, employment in modern manufacturing and service sectors is to be increased in order to achieve greater regional and international integration. The specific priorities contained in the five pillars of Vision 2050 are extremely ambitious, and include:

- competitive manufacturing anchored to a regional logistics hub;
- modern and innovative services sectors driving transformative growth;
- export-oriented knowledge services;
- high-end sustainable tourism;
- a knowledge-intensive economy;
- modern, market-oriented and climate-resilient agriculture;
- scaled-up use of modern inputs and technologies to maximise productivity;
- integration within global value chains for higher-value products;
- ease of mobility and efficient transport;
- smart and green cities for sustainable growth; and
- sustainable supply and demand for energy.

Vision 2050 is the guiding principle for strategies such as the National Strategy for Transformation and the 'Made in Rwanda' policy, through which the Rwandan government is promoting skills development, job creation and domestic value creation in sectors such as agri-tech, cybersecurity, information technology, industry, and engineering fields such as aviation.

The Rwandan government recognises technical and vocational education and training (TVET) as one of the central pillars of economic transformation and industrial development.

However, Vision 2050 also makes it clear that its targets cannot be achieved with a conventional approach: **"Business-as-usual" is not sufficient to reach the objectives of Vision 2050**' (Republic of Rwanda, Vision 2050, Abridged Version, p. 13).

This applies not least to **technical vocational education and training (TVET)**. In order to be able to fulfil the ambitious priorities mentioned, the country needs - not only but also - highly trained specialists and technicians in the middle qualification range (levels 3 to 5).

One way to achieve this is to establish so-called **TVET Centres of Excellence, a path that numerous countries have already taken** in order to achieve the **highest quality of training** in selected sectors.

Rwandan TVET Centres of Excellence will focus on strengthening existing education and training programmes and integrating new education and training programmes into the districts' development plans. The TVET Centres of Excellence will serve as **role models** for the rest of the Technical Secondary Schools (TSSs) in the district.

The construction of several new TVET Centres of Excellence and the expansion of the capacity of another 20 to 25 centres are also expected to contribute to the ambitious target of 60% TVET students in secondary education (Vision 2050). In 2019, the share of TVET students in secondary education was around 32%.

Rwandan TVET Centres of Excellence

TVET Centres of Excellence are former Technical Secondary Schools (TSSs) or newly established VET providers that collaborate with economic, educational, social and political partners and aim to provide **high-quality TVET skills** and to contribute to **regional development planning and growth**.

Their excellence can be described in terms of the:

■ Extended scope of their activities

Rwandan Centres of Excellence will not only serve students at levels 3 to 5, but also other target groups, such as:

- students at other TSSs, especially for the practical parts of levels 3 to 5 training;
- trainers at other TSSs, especially for training in their core technical subject;
- graduates of polytechnics and universities for specific practical components;
- in-company trainers;
- employees of companies and enterprises;
- any other sector specific skills upgrading short courses on various modern technologies; and
- the provision of specific training content with rare specialist skills in modern technology such as 3D printing, the IoT, robotics, AI, drone control, smart farming, and other emerging technologies;

■ High quality of their performance and their capacity to respond and develop to meet the needs of their stakeholders

Rwandan Centres of Excellence will provide infrastructure, technology, learning accommodation and human resources that reflect the skills requirements of:

- Rwandan companies operating in their home market in sectors in which Rwanda wants to **position itself internationally**, such as the IT, manufacturing, agriculture, hospitality and tourism, and logistics sectors;
- Rwandan companies that want to **export products and services** for which **international quality standards** apply, such as in the food or IT industry;
- **international companies investing in Rwanda**, for instance in the pharmaceutical industry or the logistics sector; and
- **international labour markets** to which Rwandan professionals wish to migrate (e.g. Europe, Middle East, USA).

■ Leadership role in the national or regional VET system

Rwandan Centres of Excellence will act as models, coordinators or pathfinders for other TSSs in the following aspects:

- modern training facilities and technology;
- quality delivery of education and training;
- competence-based training (CBT) implementation at levels 3 to 5;
- competence-based assessment (CBA) implementation at levels 3 to 5;
- skills assessment against international standards for selected trades;
- workplace learning;
- career guidance services;
- production units and income-generating activities;
- applied research and innovation;
- recognition of prior learning.

Rwanda TVET Centres of Excellence: services

1 High-quality teaching and learning at levels 3 to 5

TVET Centres of Excellence to reflect industry needs

Many Technical Secondary Schools (TSSs) in Rwanda are still struggling with basic quality problems such as trainers' competencies, trainers' technical English, trainers' IT skills, and a lack of textbooks and modern equipment, in particular for practical training.

TVET Centres of Excellence will take a strategic approach to ensuring that resources are tailored to the skills needs of industries and professions, in line with current international technological developments. They will work with employers and professional bodies to meet resourcing

needs. Infrastructure, technology, learning accommodation and other resources will meet industry and professional needs.

- In concrete terms, this means that the future TVET Centres of Excellence would meet the requirements and frameworks set for competence-based training (CBT). These requirements and frameworks are found, among others, in the training organisation guides, which include specifications for equipment, tools, workshops, trainer profiles, etc.

Key quality standards for industry-driven skills development

- TVET Centres of Excellence have formal arrangements in place for ensuring that social-partner representatives are involved in their decision-making bodies (such as Centres' boards, advisory councils, programme committees and curricula commissions).
- The Centres systematically consult with different stakeholders in the region so that their skills provision and activities can respond to their needs.
- A wide range of tools to generate labour-market data are used when designing and reviewing programmes.
- Graduate tracking surveys are conducted to inform the design and review of programmes.
- Employer satisfaction surveys are conducted to inform the design and review of programmes.
- The Centres have the necessary equipment and infrastructure to make use of digital technologies in teaching, learning and assessment.
- The programmes offered by Centres will systematically support (as a cross-cutting issue) the development of entrepreneurship skills and an entrepreneurial mindset.
- There is a system in place that ensures the continuous professional development of trainers.

A selection of possible performance indicators

- % of graduates finding employment within 1 year in a sector corresponding to their training.
- % of graduates indicating that the skills obtained helped them to find employment.
- % of graduates indicating that the skills obtained helped them in their job performance.
- % of recent graduates making use of the Centre's support services for starting their own business.
- % of employers indicating that the skills provided are relevant to their needs.
- Overall drop-out rate.
- % of trainers recently participating in training activities related to their pedagogical and teaching or specialist vocational skills.
- Number of other skills providers (e.g. TSSs in the same district) or other educational organisations with which a Centre has carried out joint activities lasting at least 12 months.

TVET Centres of Excellence to support workplace learning

In Rwanda, the predominant type of workplace learning in level 3-5 programmes is the Industrial Attachment Programme. A 1-month attachment is planned in all 3 years. However, public and private companies are unable to provide enough attachments, leaving many learners unserved. Another key problem is the quality of attachments. Especially in the agricultural sector, many of the predominant micro- and small enterprises cannot provide opportunities for quality learning. The quantitative need for workplace learning opportunities in Rwanda is enormous, and will increase in the future due to the

policy to significantly increase the proportion of learners in formal TVET (target of 60%). Companies will be unable to meet this demand for the foreseeable future.

- Well-equipped TVET Centres of Excellence (or their production units; see section 3.4) will systematically compensate for missing opportunities of workplace learning in companies (industrial attachments, work-based learning components of dual programmes).
- TVET Centres of Excellence will make use of equipment (or production units) that is also available to learners from other TSSs in the district or country.



Key quality standards for workplace learning

- TVET Centres of Excellence will have formal arrangements in place to ensure that employers or employer organisations are systematically involved in the design, delivery and evaluation of the programmes and services offered.
- The centres will have qualified staff and systems to match learners with training companies and to plan and organise the cycle of learners between centre and company.
- The centres will have standards for in-company trainers in their respective economic sectors. They will train in-company trainers and have programmes for their professional development.

A selection of possible performance indicators

- % of learners taking part in employer-based practical training periods (industry attachment, placements, internships, dual training) during the last school year.
- % of recent graduates finding employment with the employer where they participated in employer-based practical training periods (industry attachment, placements, internships, dual training) during their programme.
- % of curriculum time organised as employer-based practical training periods (industry attachment, placements, internships, dual training) in the last school year.
- Number of current partnerships with employers that have lasted for at least 3 years.
- Number of in-company trainers trained.



2 Short courses for youth and adults

A wide range of short courses in continuing vocational education and training will be part of the offer of Rwanda TVET Centres of Excellence. In addition to their full-time initial VET programs at levels 3 to 5, the well-resourced TVET Centres of Excellence will focus on short courses tailored to the specific needs of the industry.

Such specialist training courses usually require special equipment and trainers who have extensive practical experience with the latest technologies. Since it is too expensive to maintain such resources in a large number of TSSs, the TVET Centres of Excellence will take over the responsibility for specialist continuing vocational training in fields such as advanced manufacturing and renewable energy. However, Rwanda would need several TVET Centres of Excellence in sectors that still build the backbone of the economy, such as agriculture. For each district, there will be a thorough analysis into which economic sector the TVET Centre of Excellence should focus on.

Short courses often focus on specific technical skills. TVET Centres of Excellence in the renewable energy sector, for example, offer short courses on photovoltaic solar systems, solar thermal systems, wind systems, energy efficiency, biogas, electricity, quality assurance, and health and safety at work. Courses or modules on entrepreneurship competencies will also be important components of a TVET Centre of Excellence's offering.

In Rwanda, demand for such courses will grow as industrialisation increases. Managers from the food processing industry, for example, reported that they repeatedly have to recruit skilled workers from abroad to operate, programme and maintain

manufacturing technology, partly because there is a lack of upskilling for domestic technicians. In Rwanda there are numerous public and private providers of short vocational courses, especially in the agricultural sector. It is obvious that the future TVET Centres of Excellence will position themselves at the upper end of the spectrum of short courses because of their resources (excellent trainers, equipment, facilities, etc.).

Short courses for different target groups

- TVET Centres of Excellence will develop (in collaboration with the RDB, PSF, SSCs, MINICYOUTH, MIFOTRA, MIENDUC and the RTB) and offer standardised and customised short courses for different target groups of youths and adults, including:
 - employees from industry: upskilling;
 - self-employed people;
 - students and graduates from TSSs, e.g. learners who want to add modules to their level 3-5 programme, for example in order to apply for jobs that require international certification; and
 - students and graduates from polytechnics and universities, e.g. learners who want to add modules to their level 6-9 programme, for example in order to apply for jobs that require international certification.



Short courses for staff in the TVET sector

- TVET Centres of Excellence will also offer short courses for staff in the TVET sector itself including courses for:
 - technical trainers from TSSs that need skilling or upskilling in their core technical subject;
 - in-company or enterprise trainers who supervise and train learners as part of industrial attachment or dual programmes in companies; and
 - assessors (see also section 3.3).

Additional services for all target groups include:

- training-needs assessments for short courses (in collaboration with the RDB and the RTB);
- developing teaching and learning materials for short courses;
- a variety of formats, including online courses. Online courses can be flexibly designed to meet the needs of learners who need access to training modules at flexible times and, most importantly, can choose the lessons for their skills according to their needs. For instance, entrepreneurship skills modules are particularly well suited for online courses; and
- a course application system on the website of TVET Centres of Excellence;

Career guidance and counselling for potential learners

Coupled with a diverse range of short courses is the need for sound career guidance and counselling. TVET Centres of Vocational Excellence will have a role in career guidance for potential future learners, in particular in continuous education and training. There are strong reasons why career guidance is especially important in vocational education and training: more adult learners with limited formal education will enter short courses, due to a massive need for reskilling and upskilling.

Cost-recovery contribution

TVET Centres of Excellence generate a cost-recovery contribution, in each case to be determined by the Board of Management, through courses for industry employees and self-employed people. The Centres develop a business case for this purpose.

TVET Centres of Excellence offer special boarding facilities for adult learners.



Key quality standards for short courses

- Information on all short courses is available on the TVET Centre for Excellence's website. This includes whether the short course offers an (international) certificate or recognition.
- Counselling and career guidance services regarding the offer and selection of short courses are available for learners and companies.
- Information on entry requirements (e.g. required prior knowledge, skills and competences or qualifications) for short courses is available on the website.
- The aims and content of the short courses, as well as the forms of learning and methods used (including online and blended learning and work-based learning), are explained on the website.
- Teaching and learning materials for short courses are available on the website.
- Information on the qualifications of the teaching staff and the use of equipment, workshops and laboratories is available for each short course on the website.
- The costs and possible funding opportunities (e.g. incentives) for short courses are transparent for learners and companies and available on the website.
- A course application system is available on the website.
- TVET Centres of Excellence offer special boarding facilities for adult learners.
- The unit in charge of short courses carries out regular (self-) evaluations.

A selection of possible performance indicators

- % of graduates indicating that the skills obtained helped them to find employment.
- % of graduates indicating that the skills obtained helped them in their job performance.
- % of employers indicating that the skills provided are relevant to their needs.
- % of current learners who received careers information or advice in relation to their short course before enrolling in the programme.

3 Assessment of learners

In Rwanda, TVET programmes are only accredited, and qualifications only recognised, if they have been developed in compliance with the guidelines applicable in the country. Both level 3-5 programmes and future-demand-oriented short courses follow a competence-based training (CBT) approach.

While NESAC is responsible for national exams, the TVET Centres of Excellence should be given responsibility for assessing learners in short courses in their respective economic sectors or trades. It is not uncommon for such assessments to require compliance with

international standards, for example if skilled workers are to be prepared for international labour markets and need to be certified accordingly. For example, welders in many countries require special certification.

- The RTB accredits TVET Centres of Excellence to assess learners on short courses.
- TVET Centres of Excellence assess learners on short courses. Priority is given to assessments that must meet international standards.

Key quality standards for assessment

- TVET Centres of Excellence have the appropriately qualified staff to plan and coordinate assessments, whether in their own facilities or in external ones, e.g. at companies.
- Internal and external assessors have the necessary industry qualifications and professional experience for assessing against specific qualifications (e.g. short courses) or units within qualifications.
- Assessment verification and documentation processes and systems are in place.
- The Centres use a variety of assessment approaches to validate and recognise learning outcomes from all types of learning contexts, such as skills demonstration, work-based assessment, project-based assessment, learning diaries, etc.
- The Centres provide continuous professional development to staff members responsible for the assessment of learners.

A selection of possible performance indicators

- % of training personnel who are qualified or authorised as assessors.
- Number of assessments that follow international standards.

4 Production units

Production units are places of both learning and work. They are structured along the lines of real companies, and the trainees, who are usually young learners, are remunerated for their work. Production units produce marketable products in their workshops or offer services for real customers in their work areas. The work processes must be designed in such a way that they do not impose excessive demands on the young learners or employees.

Production units can be organised in different ways.

Permanent production: filling a gap in the market

In some areas, such as the production of furniture, agricultural products or foodstuffs, production units can be operated permanently and their products can fill a gap in the market. Such production units have an organised in a comparable way to real companies, e.g. warehouse, production, maintenance, administration, sales and marketing, etc. The production and integrated training will not usually be carried out by teachers, but by technicians and engineers recruited on the market.

Singular individual orders or projects

In other areas, such as manufacturing or IT, production units can be used in some phases of training only, for example to manufacture a specific part, to develop software or to carry out a consultancy project. In these cases, trainees usually work together with their trainers on time-limited projects on concrete solutions, mainly for usually small and medium-sized enterprises. These small and medium-sized enterprises lack the latest technology found in TVET Centres of Excellence. For example, producing a metal or plastic part for a machine using a CNC machine available at such a centre.

TVET Centres of Excellence use production units to systematically compensate for the lack of workplace learning opportunities at companies (industrial attachments, work-based learning components of dual programmes).

TVET Centres of Excellence also make use of the production units available to learners from other TVET schools or regions.

Cost-recovery contribution

Production units generate a cost-recovery contribution.





Key quality standards for production units

Production units are structured like companies. They cover order acquisition, material procurement, preparation of cost estimates, production and delivery, and marketing.

- Production units produce marketable products and services.
- Production units receive orders from external customers.
- Functional and suitable machines, tools and health and safety equipment are available in the workshops for production.
- Production units have a teaching concept and a curriculum that enable the linking of learning and work processes.
- Learners in the production units are selected according to transparent procedures.
- The working hours, remuneration, etc. of learners are clearly regulated.
- Production units maintain close cooperative relationships with other stakeholders, e.g. RTB, local administration, business organisations and individual companies, other TSSs, polytechnics etc.
- The trainers, managers, etc. working in production units have the necessary teaching, technical and business management skills.
- The trainers, managers, etc. working in production units receive regular further training.
- Production units carry out regular self-evaluations.

A selection of possible performance indicators

- % of graduates indicating that the skills obtained helped them to find employment.
- % of graduates indicating that the skills obtained helped them in their job performance.
- % of employers indicating that the services and/or products provided by the production unit are relevant to their needs.
- Number of new products and services developed by the production unit.
- Income generated by the production unit.

5 Applied research and Innovation

Applied research activities and services conducted by TVET Centres of Excellence have a strong practical focus on solving real-life problems. Findings from applied research can be used to develop or improve products and services by commercialising ideas that may increase business productivity.

An important reason for the growing role of TVET Centres of Excellence in delivering applied research is the growing need for micro, small and medium-sized enterprises to seek support for their innovation activities from nearby – but external – stakeholders.

The applied research of the TVET Centres of Excellence may have various customers or clients:

- A TVET Centre of Excellence may conduct applied research in order to develop or further develop its own products. One example would be research in the field of precision agriculture, e.g. using drones to monitor and manage crops in difficult terrain such as in Rwanda. Another example would be developing simple household appliances such as toasters or kettles, which are used in many households and hotels in Rwanda but have so far been imported.
- A TVET Centre of Excellence may conduct applied research on behalf of external commercial customers. These can be small-scale contracts for applied research, e.g. from manufacturing companies, food producers or farmers. An example would be developing environmentally friendly packaging for dried fruits. In general, there is a high demand for services such as food processing and packaging, quality assurance of food, and procurement of market information, especially from small and medium-sized enterprises.
- A TVET Centre of Excellence may conduct applied research in close collaboration with non-commercial stakeholders such as municipalities, municipal cooperatives or NGOs. An example from the agricultural sector would be the improvement of cultivation techniques under specific local conditions.



Key quality standards for applied research and innovation

- TVET Centres of Excellence cooperate in the field of applied research with other research institutions, such as polytechnics or universities.
- Applied research at TVET Centres of Excellence aim to solve practical problems. The Centres therefore work closely with industry, companies and other stakeholders to identify relevant research questions and develop solutions.
- TVET Centres of Excellence have suitably qualified staff to carry out applied research.
- The methods of applied research comply with scientific standards. This includes the use of suitable methods, the validation of results, and transparency in the research work.
- Mechanisms are in place to put research results into practice. This can be done through technology transfer, patents, cooperations with companies, cooperatives, municipalities, etc.
- A concept is in place to ensure that ethical guidelines are adhered to and that the applied research has a positive long-term impact.

A selection of possible performance indicators

- Overall number of a centre's current partnerships with enterprises or research organisations.
- % of trainers participating in innovation or research activities with partners in the last school year.
- % of learners participating in innovation or research activities with partners in the last school year.

Intake capacity of Rwandan TVET Centres of Excellence

The average size of a Rwandan TSS is relatively small. Most schools have fewer than 500 students. The ideal size of a vocational school is between 1 200 and 2 000 students. This number of students is usually considered the optimum in terms of manageability and learning environment.

In each vocational field – e.g. construction, agriculture, etc. – a TVET Centre of Excellence should have two or three parallel courses/ classes with 20 to 30 students. Only in this way is it possible to have a larger teaching staff that enables a professional departmental structure and can also compensate for sickness, etc. Furthermore this would allow for:

- more efficient and effective professional development of teachers and trainers, for example through internal training, peer learning, etc.;
- efficient investment in workshops, laboratories, equipment and teaching and learning materials; and
- specialisation and differentiation of vocational training programmes, by, for example, offering different specialisations – e.g. in the final year – after a common foundation programme in year 1 or 2.

A stronger specialisation of vocational schools in one or more vocational fields would increase economies-of-scale efficiencies, especially in investment-intensive areas such as automotive, agri-technology, food technology, industrial automation technology, etc.

TVET Centres of Excellence need to provide adequate dormitories in order to take in additional students from remote areas and to allow for broad access to capital-intensive TVET programmes.





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